CS 5330 HW9

Questions

1. A data set is usually divided into three parts: training, validation, and testing. What is the purpose, or use, of each of the three parts of the data set?

   Training is the actual data that the model learns from.

   The validation set is used to steer the training process, from determining when enough training is sufficient, choosing hyperparameters, etc.

   The testing set is used to evaluate the model after training

2. How is cross-validation different from having separate training and test sets? Why would you use cross-validation?

   cross-validation rotates what data is used for both training and evaluation. It is useful because it gives a more reliable performance estimation

3. What are the factors that enabled the deep network revolution?

   The discovery that GPUs can be used for training large models efficiently, and the subsequent development of algorithms that take advantage of GPU hardware.

4. What are the definitions of bias and variance, with respect to ML models? How are the measures used to guide development of an ML system?

   Bias is the model's error on the training set which it was trained on. Variance is the model's error on the testing set minus the bias. If you know your system is performing poorly by a specific metric, you can know how to fix it. For example, an issue with high variance can be solved with more data, regularization, or a simpler model, while an issue with high bias can be solved with a more powerful model.

5. What two ML innovations did the creators of the ALVINN network pioneer?

   The confidence estimate and data augmentation

6. What are two methods of regularizing deep networks?

   The dropout layer and the normalization layer

7. Consider AlexNet (you can find the paper on Files)
    a. How many values are there at the end of the convolution stack (last pooling layer)?

       The pooling layer brings it down to 6×6×128 ×2 = 9,216

    b. How many weights connect the end of the convolution stack with the first linear layer?

       Since the connection is dense, there are 9,216 × 2,048 × 2 weights = 37,748,736 weights connecting here.

    c. How much is the network compressing the information in the original image from the input image to the second linear layer (what you would likely use as the embedding)?

       150,528 / 4,096 = 36.75 times compression

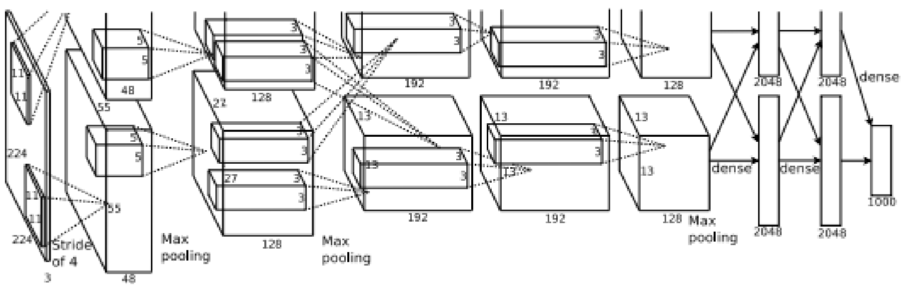

Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.